
MA 704: Dynamical System

Course Code	MA 704
Course Name	Dynamical System
L-T-P-C	3-0-0-3
Prerequisites	Knowledge of Functional Analysis, Ordinary and Partial Differential Equations, Linear Algebra
Intended for	M.S./Ph.D.
Distribution	Elective
One-time approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Linear Systems**

Diagonalization, Exponentials of Operators, Fundamental theorem for linear systems, Jordan Forms, Stability Theory, Nonhomogeneous linear systems. [10 Lectures]

- **Local Theory of Nonlinear Systems**

Existence Uniqueness Theorem, Maximal Interval of Existence, Flow, Stable Manifold Theorem, Hartman-Grobman Theorem, Lyapunov Functions, Nonhyperbolic Fixed Points, Centre Manifold Theorem, Normal Form Theory, Gradient and Hamiltonian Systems. [15 Lectures]

- **Global Theory of Nonlinear Systems**

Global Existence Theorem, Periodic Orbits, Limit Cycles and Separatrix cycles, Poincare Map, Stable Manifold Theorem for Periodic Orbits, Poincare-Bendixson theory, Lienard Systems, Bendixson's criteria, Poincare Sphere and Behaviour at Infinity, Global Phase Potraits and Separatrix Configurations, Index Theory. [15 Lectures]

Text Books

1. A.A. Andronov, E.A. Leontovich, I.I. Gordon, and A.G.Maier, **Qualitative Theory of Second Order Dynamical Systems**, John Wiley.
2. V.V. Nemytskii and V.V.Stepanov, **Qualitative Theory of Differential Equations**, Princeton University Press.

References

1. S.Lefschetz, **Differential Equations: Geometric Theory**, Interscience.
2. J.Palais and W. De Melo, **Geometric Theory of Dynamical Systems**, Springer Verlag.